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5 ways data centers endanger their local communities and the country as a whole

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8–10 minutes

Every internet search, streamed video and AI-generated response depends on a data center somewhere. Driven by rapid growth in artificial intelligence, cloud computing and cryptocurrency, data centers have become the backbone of the modern digital economy. But though their key role is in enabling virtual and remote experiences, data centers are physical buildings in real communities around the nation and the globe.

The United States hosts [more than 4,000 data centers](#) – [more than any other country](#). The U.S. Department of Energy expects that, taken together, all U.S. data centers will [consume as much as 12% of all U.S. electricity by 2028](#). In 2023, data centers consumed [about 4.4% of total U.S. electricity](#) – roughly 176 terawatt-hours.

In the U.S., Virginia has more data centers than any other state – over 600, [two-thirds of which are in the northern Virginia](#) suburbs of Washington, D.C. In 2023, the state's data centers consumed [about 26%](#) of Virginia's total electricity supply – a higher share

than in any other state.

We study [science communication](#), [climate science](#) and [public health](#), so we wanted to understand how data centers in Virginia [affect the people who live near them](#) and the broader public.

We found that the data centers that already exist affect nearby residents and the nation as a whole in five main areas: air quality, water quality, noise levels, land use and energy costs.



A row of generators sits alongside a data center building in northern Virginia.

[Andrew Caballero-Reynolds/AFP via Getty Images](#)

Air pollution

Data centers generally operate 24/7 and consume enormous amounts of electricity, which must be generated somewhere – either near the data center or farther away.

When fossil fuels are burned to generate that power, they [emit a](#)

[wide range of air pollutants](#), including those linked to lung disease, cardiovascular disease, stroke and neurological conditions. They also emit [heat-trapping pollution that causes global warming](#) and climate change, which, in turn, [worsens air pollution](#) further.

Generating power for U.S. data centers in 2023 emitted the [equivalent of 2.2%](#) of the nation's greenhouse gas emissions.

Other air pollutants emitted from fossil-fuel combustion are associated with increased risk of [ADHD](#) and [autism](#) in children and risks of [Parkinson's and Alzheimer's](#) diseases in older adults.

Unless the energy powering data centers comes from clean energy sources, such as solar, wind or geothermal, generating that electricity also pollutes the air. People who live near fossil-fuel burning power plants, whether in communities that also host data centers or in distant states, are [exposed to air pollution](#). And during electrical outages, [on-site diesel generators](#) kick in, releasing large amounts of air pollution that [can harm data center employees and nearby residents](#) alike.

Water consumption and pollution

Data centers require [vast quantities of water](#) to cool their servers. Globally, they are projected to consume between [4.2 billion and 6.6 billion cubic meters](#) of water annually by 2027. In the United States, data centers already rank among the [top 10](#) industrial water users.

In northern Virginia, data center water use has risen sharply. In Loudoun County alone, just northwest of D.C., potable water use by data centers [more than doubled between 2019 and 2023](#), while facilities across northern Virginia consumed [nearly 2 billion gallons](#)

[of water in 2023.](#)

This demand can strain [local rivers, aquifers and municipal water systems](#), even in regions like the mid-Atlantic that are not usually prone to drought, but especially in regions like the U.S. Southwest that face persistent droughts.



Cooling fans on data center roofs can spread noise and vibrations. [Andrew Caballero-Reynolds/AFP via Getty Images](#)

Noise pollution

Data centers' [continuous operation means that cooling systems, including air chillers and cooling fans](#), generate a persistent humming sound around the clock – as do any generators that are in use to provide power.

In northern Virginia, some residents have complained about an industrial-scale “drone” or “hum.” Measurements at the data

centers that were the subject of complaints found noise levels were between [40 and 59 decibels](#) on residential property.

Those noise levels are [quieter than a conversation with someone 3 feet away](#) and not loud enough to damage people's hearing or violate local noise ordinances. But they are close to levels the EPA says [reduce people's ability to work, sleep and exercise](#). Some people have complained that [data center noise has given them trouble sleeping](#) and concentrating, and some have said they avoid using their homes' outdoor spaces, where the noise is louder.



A data center under construction in Ashburn, Va., in November 2025. [Andrew Caballero-Reynolds/AFP via Getty Images](#)

Land use and community well-being

Data center expansion often targets land near [green spaces](#), [agricultural areas or rural communities](#) where developers can

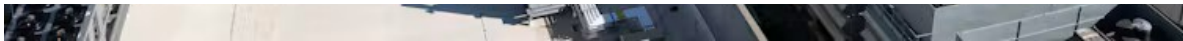
secure affordable land with access to existing electricity supplies.

Converting green space into industrial facilities can diminish [health benefits](#) associated with being in and near natural environments, including opportunities for physical activity and [improved mental well-being](#).

In Virginia, residents living near data center construction have reported [increased exposure to truck traffic and diesel exhaust](#), which can contribute to respiratory and cardiovascular health risks, especially in children and older adults. While these effects are typical of large construction projects, they can be amplified when several data centers are clustered together.

In places like Prince William County, Virginia, developers have proposed data centers on [roughly 2,400 acres](#) of undeveloped land in the Rural Crescent, an area designated by the county's planners to [remain relatively undeveloped](#). Those data centers could transform open space and rural farmland into industrial zones, disrupting communities with [long-standing ties to the land](#).





Data centers use large amounts of electricity. [Andrew Caballero-Reynolds/AFP via Getty Images](#)

Rising energy costs

As data centers increase electricity demand, they put upward pressure on energy prices across the grid. A 2024 Virginia legislative report found that the state's [typical residential electricity bill could rise by \\$14 to \\$37 per month](#) by 2040 because of grid strain tied to data center growth – a 9% to 25% increase over current average bills, and a figure that does not factor in potential inflation.

These higher costs are paid by all consumers, but they place a greater burden on families that are most economically distressed, who also tend to have more [health problems](#). Lower-income families spend a higher share of their budget on electricity, and when bills rise, the consequences can include reduced access to adequate heating and cooling, increased risks of [heat-related illness and cold-related cardiovascular stress](#), as well as difficult choices between paying for energy and food or healthcare.

What can be done

Many of these health harms can be mitigated through better planning and design.

Increasing the share of renewable energy used to power data centers would help reduce air pollution and associated health harms.

Using recycled water in [targeted systems that cool individual](#)

[server rows](#) or racks rather than whole buildings can significantly reduce cooling energy demand, with some studies estimating reductions of [up to 29%](#).

On noise, a Leesburg, Virginia, data center [reduced low-frequency tonal noise](#) by reengineering its fan mounts.

And on energy costs, requiring large-scale data centers to [cover more of the grid costs they create](#) could help protect residential customers from higher electricity bills.

The world's digital infrastructure runs through data centers, and that is not changing. We believe that expanding this infrastructure without protecting the health of surrounding communities is an unacceptable option.